MINOR PROJECT SYNOPSIS

Project ID: BTP2CSE120

Title: NETWORK PACKET ANALYSIS USING PYTHON

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✍️ 1. Introduction

With the rapid growth of internet usage, monitoring network traffic has become essential for ensuring security and performance. Every communication over a network is broken into packets. Analyzing these packets helps in understanding network behavior and detecting potential threats.

This project aims to develop a Python-based tool that captures and analyzes network packets in real time and provides useful insights into traffic patterns and anomalies.

✍️ 2. Problem Statement

Modern networks generate large volumes of packet data that are difficult to monitor manually, while existing tools are complex or expensive. A simple Python-based solution is required to capture and analyze network traffic for understanding behavior and detecting anomalies.

✍️ 3. Objectives

- To capture real-time network packets
- To analyze packet structure and protocols
- To identify suspicious traffic patterns
- To visualize network data
- To develop a lightweight monitoring system

✍️ 4. Methodology

1. Capture packets using Python libraries
2. Extract important data such as IP, port, protocol
3. Store and organize data
4. Analyze traffic patterns
5. Detect anomalies
6. Visualize results using graphs
7. Display output

5. System Architecture Design

Explanation

The system captures packets from the network, processes and extracts useful information, analyzes patterns, and finally presents results in graphical and readable formats.

🔄 6. Flowchart

START

|

Capture Packets

|

Extract Info

(IP, Port, Protocol)

|

Store Data

|

Analyze Traffic

|

Detect Anomalies

|

Visualize

|

Output

|

END

✍️ 7. Tools & Technologies

- Programming Language: Python
- Libraries: Scapy, PyShark, Pandas, Matplotlib
- Platform: Windows/Linux
- Concepts: Networking, Cybersecurity, Data Analysis

✍️ 8. Expected Outcomes

- Working packet analysis system
- Graphical representation of traffic
- Detection of abnormal behavior
- Better understanding of network protocols

✍️ 9. Applications

- Network monitoring
- Cybersecurity analysis
- Educational purposes
- Small-scale intrusion detection

✍️ 10. Conclusion

This project provides a simple and effective way to analyze network packets using Python. It enhances understanding of network communication and helps in detecting anomalies, making it useful for both learning and practical applications.